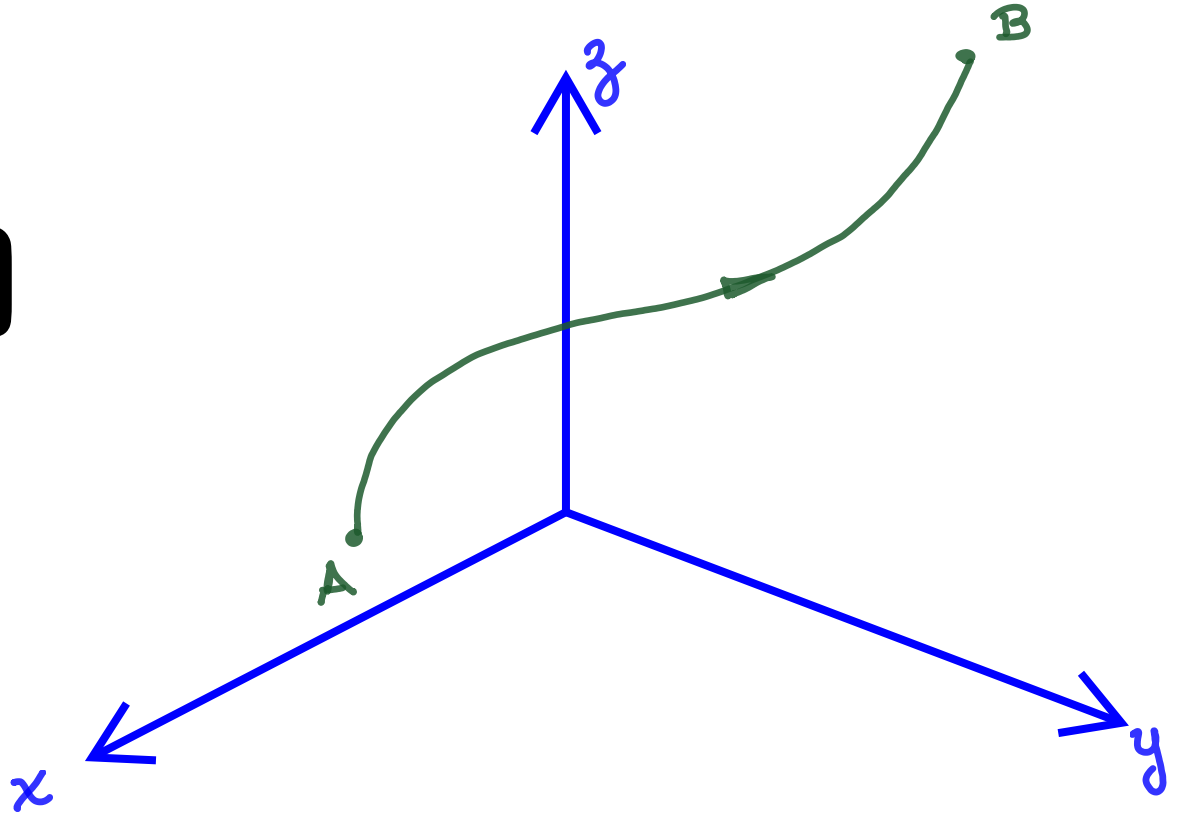


Lines in 3D

Trim Ch 11, S. 11.5

Outline:

1. Vector Functions
2. Lines in 3D
3. Forms of the equation of a line in 3D

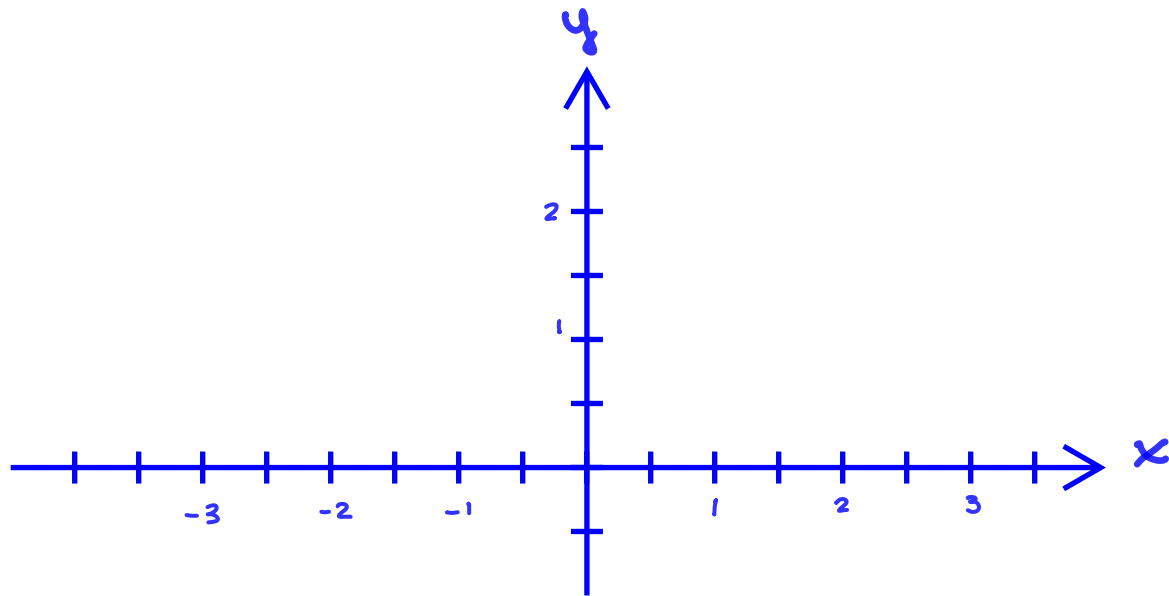


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Vector function

A vector function is a function that takes one or more variables and returns a vector!

Example: Graphing a vector function in $\mathbb{R}^2$



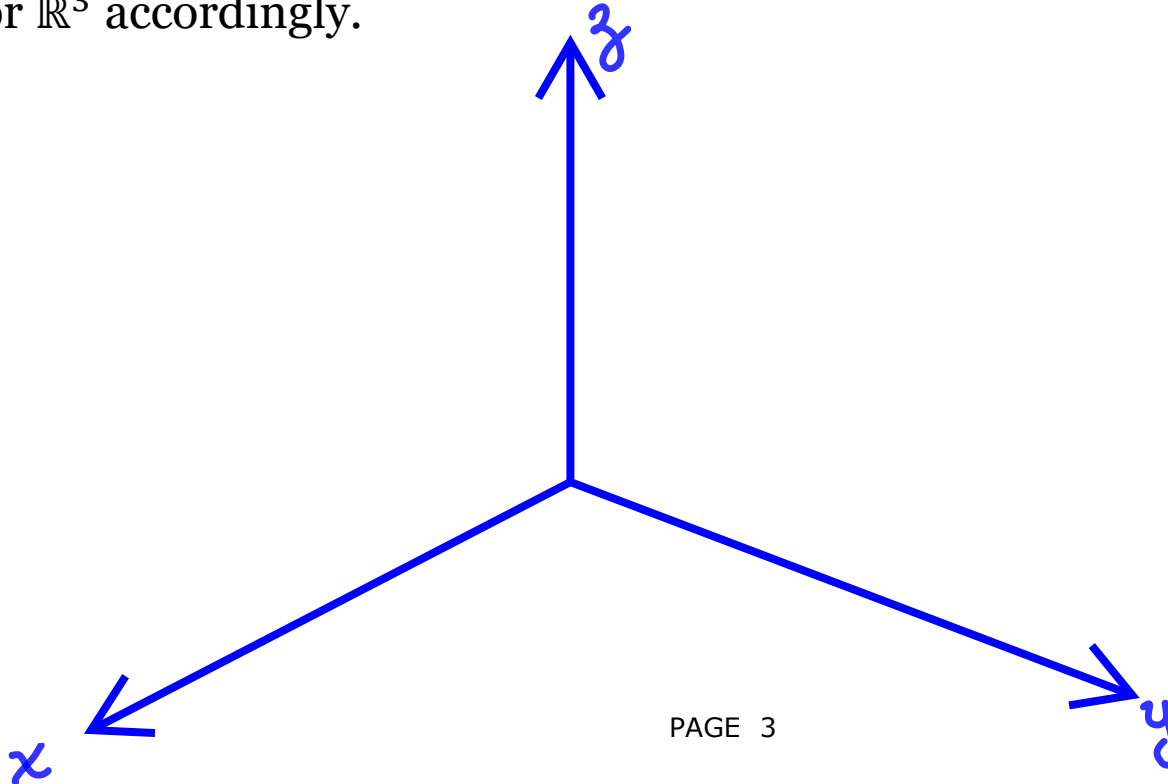
In general, a curve in $\mathbb{R}^2$ can be represented by the vector function

$$\vec{r}(t) = (f(t), g(t))$$

Extending it, a curve in $\mathbb{R}^3$ can be represented by the vector function

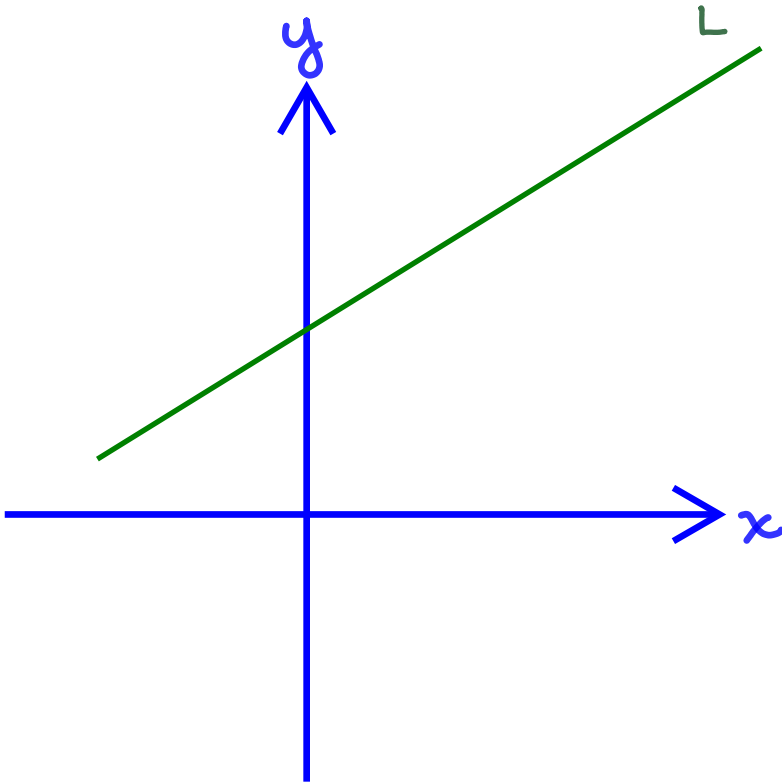
$$\vec{r}(t) = (f(t), g(t), h(t)), \quad t_A \leq t \leq t_B$$

Where $f(t)$, $g(t)$, and $h(t)$ are the coordinate components of the vectors returned in $\mathbb{R}^2$ or $\mathbb{R}^3$ accordingly.



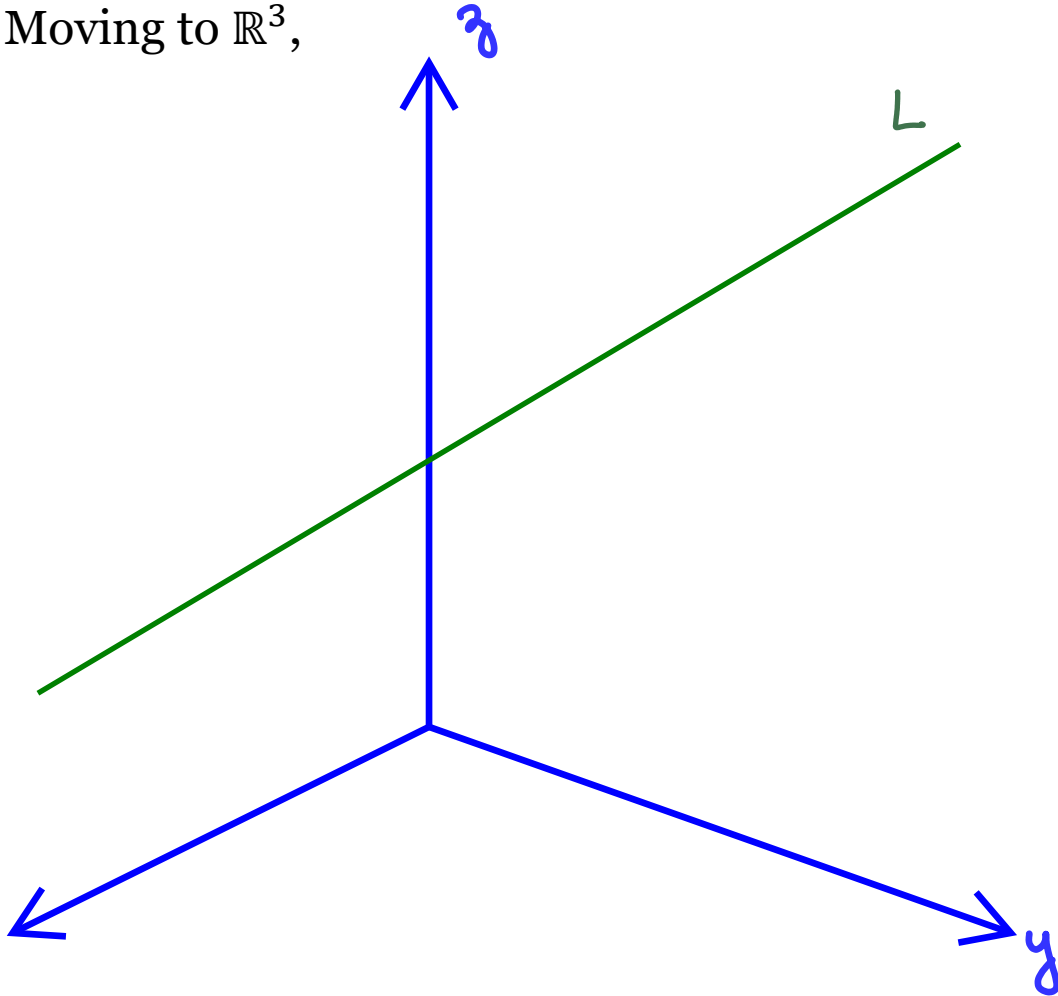
Vector equation of a line segment in $\mathbb{R}^3$

The equation of a line in $\mathbb{R}^2$ was defined by,

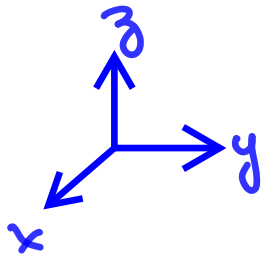
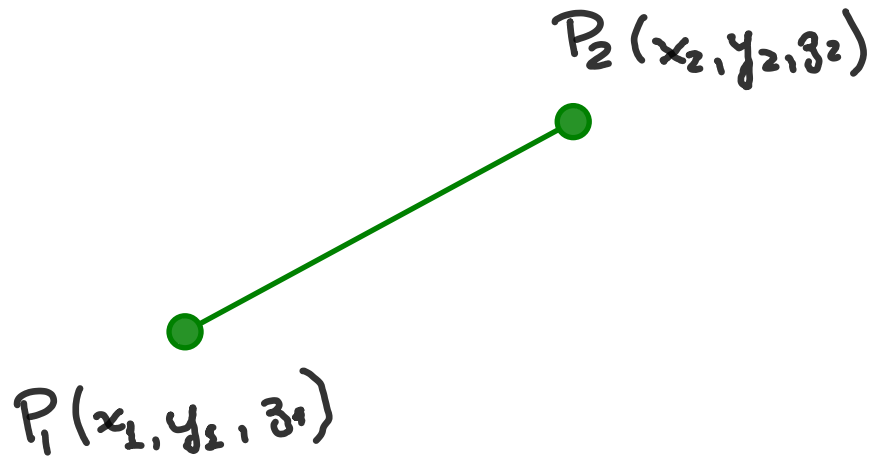


Vector equation of a line segment in $\mathbb{R}^3$

Moving to $\mathbb{R}^3$,



Vector equation of a line segment in $\mathbb{R}^3$



Vector equation of a line segment in $\mathbb{R}^3$

Other forms of the equation of a line in $\mathbb{R}^3$

Let us define the other forms of the equation of a line using an example,

Find the **vector**, **parametric** and **symmetric** equations of the line that passes through the points $A(2,4,-3)$ and $B(3,-1,1)$.

Solution:

We found that the **vector form** of the equation of a line is given by,

$$\vec{r} = \vec{r}_0 + t \vec{v}$$



Other forms of the equation of a line in $\mathbb{R}^3$

Parametric form of a line:

We find the parametric form of the line by expressing the vector equation of the line, $\vec{r} = \vec{r}_0 + t\vec{v}$, as follows,

$$(x, y, z) = (x_0, y_0, z_0) + t(a, b, c)$$

Other forms of the equation of a line in $\mathbb{R}^3$

Symmetric form of a Line:

From the parametric form of the line derived previously, and assuming that $a \neq 0$, $b \neq 0$, and $c \neq 0$, we solve each equation for “ t ”

The symmetric form of a line is then written as,

In our example,

Summary

- We defined a vector function for the equation of a line in 3D
- We represented the equation of a 3D line in three forms:
 - Vector form
 - Parametric form
 - Symmetric form
- With some practice you should be able to write the equation of a line in 3D in all its three forms



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Thank You